



## Dry Bean

Donated on 9/13/2020

Images of 13,611 grains of 7 different registered dry beans were taken with a high-resolution camera. A total of 16 features; 12 dimensions and 4 shape forms, were obtained...

### Dataset Characteristics

Multivariate

### Subject Area

Biology

### Associated Tasks

Classification

### Feature Type

Integer, Real

### # Instances

13611

### # Features

16

## Dataset Information



### Additional Information

Seven different types of dry beans were used in this research, taking into account the features such as form, shape, type, and structure by the market situation. A computer vision system was developed to distinguish seven different registered varieties of dry ...

### Has Missing Values?

No

## Introductory Paper



### [Multiclass classification of dry beans using computer vision and machine learning techniques](#)

By M. Koklu, Ilker Ali Özkan. 2020

Published in Computers and Electronics in Agriculture



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| Variable Name   | Role    | Type       | Description                                                                                    | Units  | Missing Values |
|-----------------|---------|------------|------------------------------------------------------------------------------------------------|--------|----------------|
| Area            | Feature | Integer    | The area of a bean zone and the number of pixels within its boundaries                         | pixels | no             |
| Perimeter       | Feature | Continuous | Bean circumference is defined as the length of its border.                                     |        | no             |
| MajorAxisLength | Feature | Continuous | The distance between the ends of the longest line that can be drawn from a bean                |        | no             |
| MinorAxisLength | Feature | Continuous | The longest line that can be drawn from the bean while standing perpendicular to the main axis |        | no             |
| AspectRatio     | Feature | Continuous | Defines the relationship between MajorAxisLength and MinorAxisLength                           |        | no             |
| Eccentricity    | Feature | Continuous | Eccentricity of the ellipse having the same moments as the region                              |        | no             |



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| Variable Name | Role    | Type       | Description                                                                                   | Units | Missing Values |
|---------------|---------|------------|-----------------------------------------------------------------------------------------------|-------|----------------|
| ConvexArea    | Feature | Integer    | Number of pixels in the smallest convex polygon that can contain the area of a bean seed      |       | no             |
| EquivDiameter | Feature | Continuous | Equivalent diameter: The diameter of a circle having the same area as a bean seed area        |       | no             |
| Extent        | Feature | Continuous | The ratio of the pixels in the bounding box to the bean area                                  |       | no             |
| Solidity      | Feature | Continuous | Also known as convexity. The ratio of the pixels in the convex shell to those found in beans. |       | no             |

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Additional Variable Information

- 1.) Area (A): The area of a bean zone and the number of pixels within its boundaries.
- 2.) Perimeter (P): Bean circumference is defined as the length of its border.
- 3.) Major axis length (L): The distance between the ends of the longest line that can be ...

Class Labels

Seker, Barbunya, Bombay, Cali, Dermosan, Horoz, and Sira



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| File                                 | Size   |
|--------------------------------------|--------|
| DryBeanDataset/Dry_Bean_Dataset.arff | 3.6 MB |
| DryBeanDataset/Dry_Bean_Dataset.xlsx | 2.9 MB |
| DryBeanDataset/Dry_Bean_Dataset.txt  | 3.3 KB |

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## Keywords

image processing

## DOI

10.24432/C50S4B

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